

Interim Report

Modelling the Relationship Between Electric Vehicle Adoption, Public Charging Infrastructure, and Electricity-Grid Stress Across Indian States

Suunil Dabral

sdabral@mail.walshcollege.edu

Walsh College

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Mentor: Vikas S

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GitHub Repository:

<https://github.com/Suunil1977/QM640-EV-Grid-Capstone>

Introduction

Background and Context

India's electricity sector is undergoing a structural transformation driven simultaneously by two forces: the rapid scaling of electric vehicle (EV) adoption and the parallel build-out of public charging infrastructure. Between FY2019–20 and FY2025–26, EV registrations in India grew more than tenfold across states—from a negligible share of total vehicle registrations to meaningful proportions in leading states such as Goa, Kerala, and Karnataka (NITI Aayog, n.d.). This transition, while environmentally imperative, introduces a new and underexamined source of stress on the electricity distribution grid. Unlike conventional electricity demand growth, which is gradual and geographically diffuse, EV charging loads are spatially concentrated near urban areas and temporally clustered in evening hours, creating load-profile challenges that are qualitatively distinct from historical patterns (Muratori, 2018).

The Government of India has responded with policy interventions including the FAME II scheme and the PM E-DRIVE programme, which together allocated over ₹2,839 crore for public EV charging infrastructure deployment (Ministry of Heavy Industries, 2025). By August 2025, over 25,000 public charging stations had been installed across states, compared to just 12,146 operational stations as of February 2024 (Ministry of Power, 2024). Despite this rapid growth, the geographic distribution of charging infrastructure remains highly uneven: states such as Maharashtra, Delhi, and Karnataka account for a disproportionate share of total chargers, while smaller states and Union Territories have near-zero coverage (Ministry of Heavy Industries, 2025). This distributional imbalance raises the question of whether charging density itself drives EV adoption or merely reflects it—a relationship that requires empirical disentanglement.

The relationship between EV adoption, charging infrastructure, and grid demand is embedded in a broader macroeconomic and demographic context. States with higher per capita income and greater urbanisation tend to have both more EVs and better grid infrastructure, creating potential confounding in any cross-sectional analysis (Sierzechula et al., 2014). Panel data spanning multiple financial years is therefore essential for separating temporal dynamics from cross-sectional variation, and for isolating the incremental effect of EV adoption on grid stress after controlling for year-level shocks such as the COVID-19 pandemic's suppression of electricity demand in FY2020–21. This study exploits a seven-year panel of 36 Indian states and Union Territories, sourced entirely from official Government of India publications, to provide the first systematic empirical analysis of the EV–infrastructure–grid nexus at the state level in India.

The study period—FY2019–20 through FY2025–26—spans a critical inflection point in India's EV trajectory. It captures both the pre-transition baseline and the period of accelerating growth catalysed by policy incentives, making it an ideal window for detecting early-stage adoption patterns and their grid implications. The Government of India, as the primary policy stakeholder, requires empirical evidence on whether current charging infrastructure deployment is adequate, whether it is driving or lagging adoption, and whether current EV penetration rates are yet sufficient to register detectable effects on state-level grid stress indicators. This study directly addresses all three questions through four structured research questions and their corresponding hypotheses.

Problem Statement

Despite India's ambitious EV policy agenda and significant public investment in charging infrastructure, there is no systematic state-level empirical evidence on whether public charging infrastructure is associated with higher EV adoption rates, whether EV adoption is yet producing detectable increases in electricity demand growth, or whether infrastructure and adoption variables can reliably predict which states will experience high grid stress in a given financial year. This study addresses these three gaps through a panel data framework covering 36 Indian states and Union Territories over seven financial years (FY2019–20 to FY2025–26), using exclusively official Government of India data sources.

Purpose of the Study

This study aims to quantify the relationship between public EV charging infrastructure density, EV adoption, and electricity grid stress across Indian states using panel regression, group comparison, and classification methods. Specifically, it seeks to determine whether charger density predicts EV adoption (RQ1), whether high-density states have statistically higher adoption (RQ2), whether prior-year EV adoption predicts subsequent demand growth (RQ3), and whether a combination of infrastructure, adoption, and economic variables predicts the probability of high grid stress (RQ4). The findings are intended to inform evidence-based charging infrastructure policy for the Government of India.

Interim Project Status

The following summarises the current state of the project:

- **Completed:** Data collection from all six official Government of India sources; master panel construction (252 rows $\times$ 21 columns, 36 states $\times$ 7 financial years); data cleaning and structural verification (Notebook 01); exploratory data analysis (Notebook 02); all four primary statistical analyses with robustness additions (Notebooks 03–06). All six Jupyter notebooks are publicly available at <https://github.com/Suunil1977/QM640-EV-Grid-Capstone>.
- **In progress:** Interim Report preparation; GitHub repository documentation updates.
- **Pending:** Final Report incorporating policy implications and deployment recommendations; Final Presentation.

Scope and Objectives

This study is bounded to 36 Indian states and Union Territories over the period FY2019–20 to FY2025–26. Data are drawn from six official Government of India portals and Parliamentary replies. The unit of analysis is the state-financial year observation. Four research questions are examined, each with corresponding null and alternative hypotheses:

Research Question 1 (RQ1): Does public charging-infrastructure density significantly predict EV adoption across Indian states? H_0 : The regression coefficient of charger density on EV adoption share is zero ($\beta_1 = 0$). H_1 : $\beta_1 \neq 0$; charger density is significantly associated with EV share. Decision rule: t-test on charger-density coefficient and F-test at $\alpha = .05$.

Research Question 2 (RQ2): Is there a significant difference in mean EV adoption between high-charger-density and low-charger-density states? H_0 : $\mu_{high} = \mu_{low}$;

mean EV adoption is equal across groups. $H_1: \mu_{high} \neq \mu_{low}$ (two-tailed, per synopsis). Decision rule: Welch t-test at $\alpha = .05$.

Research Question 3 (RQ3): Does EV adoption significantly predict growth in state electricity demand? $H_0: \beta_1 = 0$; EV adoption has no effect on demand growth. $H_1: \beta_1 \neq 0$. Decision rule: t-test on lagged coefficient in panel regression with year fixed effects at $\alpha = .05$.

Research Question 4 (RQ4): Do charging-infrastructure, EV-adoption, and economic variables significantly predict the probability of a state being in high grid stress? $H_0: \beta_1 = \beta_2 = \dots = \beta_5 = 0$. $H_1: \exists \beta_j \neq 0$. Decision rule: likelihood-ratio χ^2 test at $\alpha = .05$.

Literature Survey

Literature Review Approach

Sources were identified through a review of peer-reviewed literature on electric vehicle adoption, charging infrastructure, and electricity grid impacts, supplemented by citation tracking from foundational works. Keyword combinations used during the review included: (1) electric vehicle adoption AND infrastructure AND panel data; (2) EV charging AND grid stress AND India; (3) network effects AND electric vehicles; and (4) logistic regression AND adoption barriers. Inclusion criteria required peer-reviewed publication in a recognised journal, relevance to at least one of the four research questions, and publication within the last fifteen years. Seminal methodological works (Cohen, 1988; Peduzzi et al., 1996) were retained regardless of publication date due to their foundational and widely cited status in statistical methodology.

Summary of Key Literature

The literature on EV adoption consistently identifies charging infrastructure availability as a critical demand-side barrier. Egbue and Long (2012), in a foundational survey study, found that range anxiety—driven by perceived infrastructure inadequacy—was among the most frequently cited barriers to EV adoption. Gulzar et al. (2024) extended this analysis to the Indian context, finding that geographic concentration of chargers in metro areas creates a two-tiered infrastructure landscape that reinforces adoption inequality across Indian states, directly informing RQ1 and RQ2.

The role of network externalities in EV markets is theorised by Li et al. (2017), who demonstrated that EV adoption exhibits positive feedback dynamics: higher adoption rates reduce per-unit charging costs, attracting further adoption. Springel (2021) provided empirical evidence from Norway showing that subsidy structures interact with network externality mechanisms to produce non-uniform geographic adoption patterns—directly relevant to India’s FAME II scheme. On the grid impact side, Clement-Nyns et al. (2010) established that uncoordinated EV charging significantly increases residential distribution grid stress, and Muratori (2018) updated this finding for the American context, showing that even at 5–10% penetration, local peak demand can increase by 25–50%. These findings motivate RQ3 and RQ4.

Sierzchula et al. (2014) found that financial incentives and per capita income are the strongest predictors of EV market share across seventeen countries, supporting the inclusion of PC-NSDP as a control variable. Singh et al. (2025), examining India specifically using panel data, found that macroeconomic variables including net state domestic product and fuel prices are significant EV adoption predictors. The methodological choices in this study are grounded in

Cohen (1988) for power analysis, Peduzzi et al. (1996) for the events-per-variable rule governing RQ4, and Faul et al. (2007) for the G*Power framework used across all four research questions.

Table 1

Literature Relevance Matrix

Author (Year)	Domain	Dataset / Setting	Method	Key Findings	RQ Linkage
Egbue & Long (2012)	EV adoption barriers	US consumer survey	Survey analysis	Range anxiety and infrastructure gap are primary adoption barriers	RQ1, RQ2
Clement-Nyns et al. (2010)	Grid impact of EVs	Belgian residential grid	Simulation	Uncoordinated charging raises peak demand on residential feeders	RQ3
Li et al. (2017)	Network effects in EV markets	US EV sales data	Structural model	Indirect network externalities amplify adoption via charging density	RQ1, RQ2
Sierzechula et al. (2014)	EV adoption determinants	17-country panel	Panel regression	Income and incentives are strongest adoption predictors	RQ1, RQ4
Muratori (2018)	EV grid impact	US residential feeders	Power systems simulation	Low EV penetration raises local peak demand significantly	RQ3, RQ4
Springel (2021)	EV subsidies and network effects	Norwegian EV market	Diff-in-differences	Subsidy design interacts with network externalities	RQ1, RQ2
Faul et al. (2007)	Power analysis methodology	G*Power software	Statistical methods	G*Power enables flexible power	All RQs

Author (Year)	Domain	Dataset / Setting	Method	Key Findings	RQ Linkage
				analysis across test families	
Cohen (1988)	Effect size benchmarks	Theoretical	Statistical methods	Established small/medium/large effect size conventions	All RQs
Peduzzi et al. (1996)	Logistic regression EPV	Simulation study	Simulation	Minimum 10 events per variable for stable logistic estimates	RQ4
Singh et al. (2025)	EV adoption drivers in India	India state panel	Panel regression	NSDP and fuel prices are significant EV predictors in India	RQ1, RQ4
Gulzar et al. (2024)	EV charging in India	India charging station data	Systematic review	Geographic concentration reinforces adoption inequality	RQ1, RQ2

Note. EPV = events per variable. RQ = research question. NSDP = net state domestic product.

Data Description

Data Sources and Access

All data used in this study are sourced from official Government of India portals and Parliamentary replies. No private, commercial, or synthesised datasets are used. Six primary sources were accessed between May and July 2026: (1) NITI Aayog India Climate and Energy Dashboard (ICED)—EV registrations by state and financial year, and statewise peak demand data (<https://iced.niti.gov.in>); (2) Ministry of Heavy Industries Parliamentary Reply PRID 2151390—state-wise installed EV charging stations as on August 1, 2025

(<https://pib.gov.in/PressReleasePage.aspx?PRID=2151390>); (3) Ministry of Power Parliamentary Reply PRID 2003003—operational charging stations as on February 2, 2024 (<https://pib.gov.in/PressReleaseIframePage.aspx?PRID=2003003>); (4) Reserve Bank of India Handbook of Statistics on Indian States 2024–25, Table 19—per capita net state domestic product at current prices (<https://rbi.org.in>); (5) Census 2011 Primary Census Abstract—urban population share by state (<https://censusindia.gov.in/nada/index.php/catalog/6191>); (6) Ministry of Health and Family Welfare Technical Group Report, November 2019, Table 8—state population projections used as the charger density denominator (https://nhm.gov.in/New_Updates_2018/Report_Population_Projection_2019.pdf). All raw files are archived in the /data/raw/ folder of the project GitHub repository.

Dataset Overview

The master analysis panel (QM640_Analysis_Panel_UPDATED.xlsx, sheet Panel_State_FY) contains 252 rows and 21 columns. The panel covers 36 Indian states and Union Territories across seven financial years (FY2019–20 to FY2025–26). The unit of analysis is the state-financial year pair. The panel structure is confirmed in Figure A1 (see Appendix A).

Table 2

Dataset Overview

Attribute	Value
Number of records (rows)	252
Number of variables (columns)	21
Time period	FY2019–20 to FY2025–26 (7 financial years)
Cross-sectional units	36 states and Union Territories of India

Attribute	Value
Unit of analysis	State $\times$ Financial Year observation
Primary outcome variables	EV share %; Peak-demand YoY growth; Grid-stress class (binary)
Primary predictor	Chargers per 100,000 population (time-invariant covariate)
Control variables	PC-NSDP at current prices (Rs); Urban % (Census 2011)
Panel type	Unbalanced panel (structural nulls for 3 UTs on peak demand)
GitHub path	data/panel/QM640_Analysis_Panel_UPDATED.xlsx

Note. PC-NSDP = per capita net state domestic product. YoY = year-on-year. UTs = Union Territories.

Data Dictionary

Table 3

Data Dictionary — Key Variables

Variable Name	Definition	Type	Allowed Values / Range	Missing Values Handling	Notes
State (ICED name)	State/UT name as labelled in ICED portal; join key	Text	Any of 36 Indian states/UTs	None (complete)	Used as panel join key across all sources
FY	Financial year in YYYY-YY format	Text	FY2019-20 to FY2025-26	None (complete)	7 financial years in panel
EV share %	EV registrations $\div$ total registrations (0–1 scale)	Float	0.000 to 0.109	None (complete)	Outcome for RQ1/RQ2; predictor for RQ3/RQ4
Public chargers (snapshot)	Operational public EV chargers per PIB annexures	Integer	0 to 2,367	Structural null — retained as null	46 nulls (18.3%): Ladakh and Mizoram absent from PIB annexures
PC-NSDP Current Rs (RBI/NSO)	Per capita NSDP at current prices (Rs)	Float	~Rs 29,000 to ~Rs 400,000	Structural null — retained as null	86 nulls (34.1%): D&NH+D&D, Lakshadweep,

Variable Name	Definition	Type	Allowed Values / Range	Missing Values Handling	Notes
					Gujarat FY2023-24, Ladakh
Urban % (Census 2011)	Urban population share from Census 2011 PCA	Float	0.0% to 97.5%	None (complete)	Complete across all 252 rows
Annual Peak Demand (MW)	Annual peak electricity demand in Megawatts	Float	0 to ~72,000 MW	Structural null — retained as null	21 nulls (8.3%): Ladakh, Lakshadweep, A&N
Peak-demand YoY growth	Year-on-year peak demand growth rate	Float	-0.163 to +0.230	Structural null — retained as null	54 nulls (21.4%): FY2019-20 + 3 UTs
Grid-stress class	Binary: 1 if YoY growth > FY median; else 0	Integer (0/1)	0 or 1	Structural null — retained as null	54 nulls; balanced: 102 class-0, 96 class-1
Chargers per 100k population	Chargers ÷ population × 100,000	Float	0.0 to 80.6	Structural null — retained as null	46 nulls matching public chargers column
EV growth (pp)	Year-on-year change in EV share % in percentage points	Float	Negative to positive	Structural null — first FY per state has no prior year	5th predictor in RQ4 per synopsis

Note. All nulls are structural — no imputation was performed. Null values are retained and excluded only at the complete-case subset construction stage for each RQ.

GitHub Data Availability Statement

The raw and processed datasets are available in the project GitHub repository at <https://github.com/Suunil1977/QM640-EV-Grid-Capstone>. Raw data: data/raw/ (four official GoI files). Processed data: data/panel/QM640_Analysis_Panel_UPDATED.xlsx. Code/notebooks: notebooks/ (six Jupyter notebooks, 01–06). Data dictionary: docs/data_dictionary.md. Source registry: docs/QM640_EV_Data_Source_Registry.md.

Screenshots and Evidence

Figure A1 in Appendix A confirms the master panel shape as (252, 21) with all 36 states, 7 financial years, and all 21 column names verified. Figure A2 confirms the null audit per column. The complete set of 21 executed notebook screenshots is provided in Appendix A, referenced throughout this report by figure number.

Analysis

Data Cleaning

Data cleaning was performed in Notebook 01 (01_data_cleaning_join.ipynb). All steps are documented in Table 4. Executed outputs are shown in Figures A1–A4 (Appendix A). No values were imputed, interpolated, or fabricated; all null values are permanent structural gaps traceable to official Government of India publications.

Table 4

Data Cleaning Log

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Structural nulls — Public chargers; charger data	Chargers per 100k (46 rows, 18.3%)	df.isnull() audit	Retained as null; excluded from RQ1/RQ2 complete-case subsets	Ladakh and Mizoram absent from both PIB charger annexures (PRID 2003003 and 2151390)
Structural nulls — PC-NSDP	PC-NSDP (86 rows, 34.1%)	df.isnull() audit	Retained as null; RQ1 complete-case N=31	D&NH+D&D, Lakshadweep, Gujarat FY2023-24, Ladakh: no NSO NSDP series published

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Structural nulls — peak demand	Annual Peak Demand; Energy Req. (21 rows, 8.3%)	df.isnull() audit	Retained as null; 3 UTs excluded from RQ3/RQ4 panels	Ladakh, Lakshadweep, A&N absent from ICED statewide exports
Structural nulls — YoY growth	Peak-demand YoY growth; Grid-stress class (54 rows, 21.4%)	Derived computation check	Retained as null; RQ3 FY2019-20 structurally uses FY2020-21 to FY2025-26 only	impossible — no FY2018-19 baseline in panel
Data type corrections	EV registrations; Total registrations (stored as float64)	df.dtypes inspection	Converted to Int64 (nullable integer)	Count variables should not be stored as floats; see Figure A3
Outlier flags	36 rows flagged (EV Z-score by FY share > 2 SD from FY mean)	year	Retained with flag; not removed	Outliers represent genuine high- adoption states (Goa, Karnataka, Kerala); removal would bias analysis

Note. D&NH+D&D = Dadra & Nagar Haveli and Daman & Diu. A&N = Andaman & Nicobar Islands. Three cross-source reconciliation gaps are declared: (1) ICED vs PIB EV totals differ by approximately 3.3%, attributable to Telangana's late Vahan 4 migration; (2) CEA All-India energy totals exceed state-row sums due to unattributed demand categories; (3) the sum of state annual peak demands exceeds the published national peak due to the coincidence factor in power systems.

EDA Results

Exploratory data analysis was conducted in Notebook 02 (02_eda.ipynb) across ten analytical sections. EV share % has a mean of 1.24% (SD = 1.77%) across all 252 observations, with strong right-skew driven by early-adopter states (see Figure A7, Appendix A). Chargers per 100,000 population has a mean of 7.23 (SD = 12.44), median of 2.4, confirming extreme geographic concentration. The Pearson correlation between chargers per 100,000 population and EV share % is $r = 0.380$ ($p < .001$), as shown in Figure A8 (Appendix A). Peak-demand year-on-

year growth has a mean of 4.9% (SD = 6.5%), with FY2020–21 showing near-zero growth due to the COVID-19 demand shock. The null value heatmap (Figure A9, Appendix A) confirms that EV share % and Urban % are the only fully complete variables across all 252 rows. The grid-stress outcome is near-perfectly balanced at 102 class-0 versus 96 class-1 observations.

Table 5

EDA Insight Summary

Figure Reference	What It Shows	Key Insight	Why It Matters (RQ Linkage)	Decision / Next Step
Figure A5–A6 (Appendix A)	Descriptive statistics — all 21 variables	EV share % right-skewed; Chargers/100k highly concentrated	Motivates regression over simple means; validates panel approach	Apply OLS; check heteroscedasticity
Figure A7 (Appendix A)	Histograms of 4 key variables	EV share and charger density strongly right-skewed	Warns against linear extrapolation at high density	Use median split for RQ2; add 3-tier ANOVA
Figure A8 (Appendix A)	Pearson correlation heatmap	Chargers/100k vs EV share $r=0.380$; moderate collinearity between predictors	Confirms positive charger-adoption association; motivates Ridge robustness	Add Ridge check for RQ1
Figure A9 (Appendix A)	Structural gap heatmap — 252 rows $\times$ 7 key columns	PC-NSDP (34.1%) largest gap; EV share % and Urban % fully complete	Justifies complete-case N=31 for RQ1; EV share % as anchor variable	Declare structural gaps; complete-case approach
FY trend analysis (Notebook 02)	FY-level mean YoY peak demand growth	FY2020-21 near-zero growth (COVID shock); FY2025-26 contraction	Year fixed effects essential for RQ3 to absorb macroeconomic shocks	Include year FE dummies in RQ3
Class balance (Notebook 02)	Grid-stress class balance	Near-perfect balance: 102 class-0 vs 96 class-1 in df_rq4	No class imbalance correction needed for RQ4	Standard logistic regression without resampling

Note. All figures are from Notebook 02 (02_eda.ipynb), available in the project GitHub repository and reproduced in Appendix A.

Sample Size per Research Question

Sample size adequacy was assessed via G*Power 3.1 (Faul et al., 2007). All four analytical subsets meet or exceed the minimum N required for 80% power. Subset sizes are confirmed by executed output in Figure A4 (Appendix A).

Table 6

Analytical Subset Sample Sizes

RQ	Subset	N	FY Coverage	Variables Required	Min N (Power)	Status
RQ1	df_cross	31 states	FY2023-24	EV share %, Chargers/100k, PC- NSDP, Urban %	25 (f^2 = 0.35, α = .05, β = .80)	PASS
RQ2	df_rq2	34 states (17 per group)	FY2023-24	EV share %, Chargers/100k	26 (d = 0.50, α = .05, β = .80)	PASS
RQ3	df_lag	165 rows	FY2021-22 to FY2025-26	EV share lag, Peak- demand YoY, year dummies	55 (f^2 = 0.15, α = .05, β = .80)	PASS
RQ4	df_cc	123 rows	FY2020-21 to FY2025-26	Grid-stress class, 5 predictors	50 (EPV $\geq$ 10, 5 predictors) = 12.0)	PASS (EPV

Note. RQ2 synopsis stated 18 states per group based on 36 total; actual data yields 17 per group from N=34 because Ladakh and Mizoram are absent from both PIB charger annexures (PRID 2003003 and PRID 2151390)—a structural data gap declared throughout this study. Power analysis conducted in G*Power 3.1 (Faul et al., 2007). EPV = events per variable (Peduzzi et al., 1996).

Modelling

Choice of Models with Justification

Four modelling approaches are deployed, one per research question, with robustness additions as described below. All primary models are hypothesis-driven; robustness additions serve as sensitivity checks and do not replace the primary hypothesis tests.

RQ1 — Multiple linear regression (OLS): EV share % is a continuous outcome variable and the research question asks about the directional and magnitude relationship between charger density and adoption controlling for income and urbanisation. OLS regression is the appropriate method for this design (Cohen, 1988). Ridge regression is added as a robustness check given moderate inter-predictor correlations. The charger density variable uses the February 2024 operational snapshot (PIB PRID 2003003), which is the snapshot assigned to the FY2023-24 cross-section in the master panel. The August 2025 snapshot (PIB PRID 2151390) is assigned to FY2025-26 rows in the panel. Executed outputs are shown in Figures A10–A13 (Appendix A).

RQ2 — Welch t-test (independent samples, two-tailed): States are split into two groups by median charger density. $H_1: \mu_{high} \neq \mu_{low}$ (two-tailed, as specified in the synopsis). Welch's t-test is preferred because it does not assume equal variances. A three-tier ANOVA robustness check and Spearman rank correlation are added as additions. Executed outputs are shown in Figures A14–A15 (Appendix A).

RQ3 — Lagged panel regression with year fixed effects: The outcome variable is peak-demand year-on-year growth (MW). The synopsis referenced energy requirement growth (MU) as the RQ3 outcome; peak demand was selected in the analysis as it offers better state-level

coverage (21 structural nulls vs 42 for energy requirement). Year fixed effects absorb macroeconomic shocks including the COVID-19 demand suppression in FY2020–21. A two-way fixed effects specification is added as a robustness check. Executed outputs are shown in Figures A16–A17 (Appendix A).

RQ4 — Binary logistic regression: Five predictors are used as specified in the synopsis: EV share %, Chargers/100k, PC-NSDP, Urban %, and EV growth (year-on-year change in EV share in percentage points). EV growth is computed as the first difference of EV share % per state to avoid division-by-zero when prior-year EV share equals zero. The model is fitted on all 123 complete-case rows (EPV = 12.0, passing the Peduzzi et al. (1996) minimum of 10). A benchmark layer of four additional classifiers plus a majority-class baseline is fitted on an 80/20 stratified train-test split for comparative evaluation only. Executed outputs are shown in Figures A18–A21 (Appendix A).

Table 7

Feature Set and Usage

Feature	Original/Engineered Type		Reason for Inclusion	Used in Model(s)
EV share %	Engineered (ratio)	Continuous (0–1)	Primary outcome for RQ1/RQ2; predictor (lagged) for RQ3; predictor for RQ4	RQ1 (Y), RQ2 (Y), RQ3 (X), RQ4 (X)
Chargers per 100k population	Engineered (ratio)	Continuous	Primary predictor of adoption (RQ1/RQ2)	RQ1 (X), RQ2 (grouping)
PC-NSDP Current Rs	Original (RBI/NSO)	Continuous	Wealth control; confounds adoption-infrastructure relationship	RQ1 (control), RQ4 (X)

Feature	Original/Engineered	Type	Reason for Inclusion	Used in Model(s)
Urban % (Census 2011)	Original (Census)	Continuous	Urbanisation control	RQ1 (control), RQ4 (X)
Peak-demand YoY growth	Engineered (rate)	Continuous	Outcome for RQ3; basis for grid-stress class derivation	RQ3 (Y)
Grid-stress class	Engineered (binary)	Binary (0/1)	Binary outcome for classification (RQ4)	RQ4 (Y)
EV_share_lag1	Engineered (lag)	Continuous	Lagged predictor to test temporal EV-to-grid effect (RQ3)	RQ3 (X)
EV growth (pp)	Engineered (first diff)	Continuous	5th predictor per synopsis; captures EV adoption velocity	RQ4 (X)

Table 8*Evaluation Metrics and Formulae*

Metric	Formula	Interpretation	Used For
R^2	$R^2 = 1 - SSR/SST$	Proportion of variance in Y explained by predictors	RQ1, RQ3
Adjusted R^2	$A = (1 - R^2)(n - 1)$ $B = n - k - 1$ $Adj R^2 = 1 - A/B$	R^2 penalised for number of predictors k	RQ1, RQ3
F-statistic	$F = MSR/MSE$	Tests whether the overall model is jointly significant	RQ1, RQ3
Welch t-statistic	$t = \frac{M1 - M2}{\sqrt{(s1^2/n1 + s2^2/n2)}}$	Tests mean difference between two groups without assuming equal variances	RQ2

Metric	Formula	Interpretation	Used For
Cohen's d	$d = (M1 - M2)/SDp$	Standardised effect size; d =0.20 small, 0.50 medium, 0.80 large	RQ2
Eta-squared (η^2)	$\eta^2 = SSB/SST$	Proportion of variance explained by group in ANOVA	RQ2 (robustness)
LR Chi-squared	$\chi^2 = -2(L0 - L1)$	Likelihood-ratio test; model vs null	RQ4
McFadden R ²	$R^2McF = 1 - L1/L0$	Pseudo R ² for logistic regression	RQ4
Accuracy	$Acc = (TP + TN)/N$	Proportion of correctly classified observations	RQ4 benchmark
Precision	$P = TP/(TP + FP)$	Of predicted positives, proportion that are true positives	RQ4 benchmark
Recall	$R = TP/(TP + FN)$	Of actual positives, proportion correctly predicted	RQ4 benchmark
F1 Score	$F1 = 2PR/(P + R)$	Harmonic mean of precision and recall	RQ4 benchmark
ROC-AUC	$AUC = \int TPR d(FPR)$	Probability that model ranks a random positive higher than a random negative; 0.5 = random	RQ4

Note. TP = true positive. TN = true negative. FP = false positive. FN = false negative. LL = log-likelihood. SS = sum of squares.

Preliminary Results

Preliminary Findings by Research Question

Research Question 1: Charger Density and EV Adoption (Multiple Linear Regression)

The OLS regression model for RQ1 (Notebook 03: 03_RQ1_regression.ipynb) was fitted on df_cross ($N = 31$ states, FY2023–24). The overall model was statistically significant ($F(3, 27) = 4.18, p = .015$), explaining 31.7% of variance in EV share % ($R^2 = .317$, adjusted $R^2 = .243$). The charger density coefficient was positive ($\beta = 0.8447$) in the hypothesised direction but did not reach individual significance ($t(27) = 1.72, p = .097$). Urban % was the strongest individual predictor ($\beta = 1.3267, p = .015$). The Ridge robustness check confirmed sign stability across all alpha values tested. The null hypothesis for RQ1 is rejected at $\alpha = .05$ based on the jointly significant F-test. Coefficient outputs are shown in Figure A10 (Appendix A).

Research Question 2: Group Comparison by Charger Density (Welch t-Test and ANOVA)

The two-tailed Welch t-test for RQ2 (Notebook 04: 04_RQ2_ttest_anova.ipynb) compared mean EV share % between high-density states ($n = 17$) and low-density states ($n = 17$) in df_rq2 ($N = 34$ states, FY2023–24). The synopsis specified 18 states per group based on 36 total; the actual data yields 17 per group from $N = 34$ because Ladakh and Mizoram are absent from both PIB charger annexures. The two-tailed Welch t-test yielded $t(31.97) = 1.499, p = .144$, failing to reject H_0 at $\alpha = .05$. Cohen's $d = 0.51$ indicates a medium effect. The three-tier ANOVA robustness check was statistically significant ($F(2, 31) = 4.81, p = .015, \eta^2 = .237$). Results are shown in Figures A14–A15 (Appendix A).

Research Question 3: Lagged EV Adoption and Demand Growth (Lagged Panel Regression)

The lagged panel regression for RQ3 (Notebook 05: 05_RQ3_lagged_panel.ipynb) was fitted on `df_lag` (N = 165 state-financial year rows, FY2021–22 to FY2025–26). The lagged EV share coefficient was near zero and statistically non-significant ($\beta = 0.0199$, $t(159) = 0.124$, $p = .902$). The overall model F-test was jointly significant ($F(5, 159) = 5.63$, $p < .001$, $R^2 = .151$), driven entirely by the year fixed effects. The two-way FE robustness check produced a consistent null result ($\beta = -0.474$, $p = .169$). The null hypothesis for RQ3 cannot be rejected. Results are shown in Figures A16–A17 (Appendix A).

Research Question 4: Predicting Grid Stress (Logistic Regression and ML Benchmark)

The logistic regression for RQ4 (Notebook 06: 06_RQ4_logistic_classification.ipynb) was fitted on `df_cc` (N = 123 complete-case rows, EPV = 12.0). Five predictors were used as specified in the synopsis: EV share %, Chargers/100k, PC-NSDP, Urban %, and EV growth. The likelihood-ratio chi-squared test was non-significant ($\chi^2(5) = 3.35$, $p = .646$), indicating that the five-predictor set does not significantly predict grid-stress class beyond the null model. McFadden $R^2 = 0.020$ confirms near-null predictive power. In-sample accuracy was 54.5% and ROC-AUC = 0.576. The ML benchmark layer found Naive Bayes to be the best-performing classifier on the held-out test set (accuracy = 60.0%, F1 = 0.600). Results are shown in Figures A18–A21 (Appendix A).

Preliminary Model Performance Comparison

Table 9

Preliminary Model Performance Comparison

Model	RQ	Validation	Primary Metric	Value	Key Takeaway
OLS + Ridge (RQ1)	RQ1	In-sample (N=31)	F(3,27)	4.18, p=.015*	Model jointly significant; charger density positive but not individually significant
Welch t-test, two-tailed	RQ2	In-sample (N=34)	Cohen's d	0.51 (medium)	Fails significance at N=34 (p=.144); ANOVA significant (p=.015)
ANOVA 3-tier (robustness)	RQ2	In-sample (N=34)	F(2,31)	4.81, p=.015*	<i>Tier differences significant; $\eta^2 = 0.237$ (large effect)</i>
Lagged OLS + Year FE	RQ3	In-sample (N=165)	β_{lag}	0.0199 (t=0.124)	Lag coefficient near zero; year FE significant; fail to reject H_0
Two-Way FE (robustness)	RQ3	In-sample (N=165)	β_{lag}	-0.474 (p=.169)	Consistent null after absorbing state-level confounders
Logistic Regression (5 pred.)	RQ4	In-sample (N=123)	LR $\chi^2(5)$	3.35 (p=.646)	Non-significant; AUC=0.576; McFadden $R^2=0.020$
Naive Bayes (best benchmark)	RQ4	Test set (N=25)	F1	0.600	Best benchmark; marginal improvement over random baseline
Majority Baseline	RQ4	Test set (N=25)	Accuracy	0.520	Reference floor for classification performance

Note. * $p < .05$. All models trained on official Government of India data from QM640_Analysis_Panel_UPDATED.xlsx.

Interim Limitations and Risks

Several limitations constrain the generalisability of these preliminary findings. First, the charger density variable is time-invariant (same snapshot repeated across all seven financial years per state), preventing detection of within-state temporal variation. Second, the PC-NSDP variable has 34.1% structural nulls, limiting the RQ1 complete-case sample to $N = 31$ states. Third, the lagged regression in RQ3 uses only a one-year lag; the true grid impact of EV

adoption may materialise over longer horizons. Fourth, the small test set ($N = 25$) in the RQ4 ML benchmark produces unreliable held-out estimates. All three cross-source reconciliation gaps—ICED vs PIB EV totals, CEA energy totals vs state-row sums, and the coincidence factor in peak demand—are declared transparently and do not affect the state-level analytical results.

Next Steps for Final Report

The following work is planned for the Final Report:

- Policy implications section translating statistical findings into actionable recommendations for the Government of India’s charging infrastructure deployment strategy.
- Workflow architecture diagram documenting the complete data pipeline from raw government sources through the six notebooks to final model outputs.
- Elasticity analysis for RQ1 quantifying the percentage change in EV adoption associated with a one-unit increase in charger density per 100,000 population.
- Extended interpretation of the null results for RQ3 and RQ4 in the context of current Indian EV penetration rates and the timeline for detectable grid impacts.

References

- Central Electricity Authority. (2025). Power supply position reports. Ministry of Power, Government of India. <https://cea.nic.in/power-supply/?lang=en>
- Clement-Nyngs, K., Haesen, E., & Driesen, J. (2010). The impact of charging plug-in hybrid electric vehicles on a residential distribution grid. *IEEE Transactions on Power Systems*, 25(1), 371–380. <https://doi.org/10.1109/TPWRS.2009.2036481>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.
- Egbue, O., & Long, S. (2012). Barriers to widespread adoption of electric vehicles: An analysis of consumer attitudes and perceptions. *Energy Policy*, 48, 717–729. <https://doi.org/10.1016/j.enpol.2012.06.009>
- Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39(2), 175–191. <https://doi.org/10.3758/BF03193146>
- Gulzar, Y., Dutta, M., Gupta, D., Juneja, S., Soomro, A. B., & Mir, M. S. (2024). Revolutionizing mobility: A comprehensive review of electric vehicle charging stations in India. *Frontiers in Sustainable Cities*, 6, 1346731. <https://doi.org/10.3389/frsc.2024.1346731>

Li, S., Tong, L., Xing, J., & Zhou, Y. (2017). The market for electric vehicles: Indirect network effects and policy design. *Journal of the Association of Environmental and Resource Economists*, 4(1), 89–133. <https://doi.org/10.1086/689702>

Ministry of Health and Family Welfare, National Commission on Population. (2019). Population projections for India and states 2011–2036: Report of the Technical Group on Population Projections. Government of India.
https://nhm.gov.in/New_Updates_2018/Report_Population_Projection_2019.pdf

Ministry of Heavy Industries. (2024). State-wise details of electric vehicles and total vehicles [Press release, PRID 2084148]. Press Information Bureau.
<https://pib.gov.in/PressReleasePage.aspx?PRID=2084148>

Ministry of Heavy Industries. (2025). State-wise details of installed electric vehicle charging stations as on 01.08.2025 [Press release, PRID 2151390]. Press Information Bureau.
<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2151390>

Ministry of Power. (2024). State-wise operational public EV charging stations as on 02.02.2024 [Press release, PRID 2003003]. Press Information Bureau.
<https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2003003>

Ministry of Power. (2025). State/UT-wise power supply position [Rajya Sabha Unstarred Question No. 150, answered 03-02-2025]. Government of India.
<https://www.powermin.gov.in/static/uploads/2025/08/5804f7f17e764f9ba1951183379b8e31.pdf>

Ministry of Statistics and Programme Implementation. (2023). Per capita net state domestic product at current prices [Press release, PRID 1942055]. Press Information Bureau.
<https://pib.gov.in/PressReleasePage.aspx?PRID=1942055>

Muratori, M. (2018). Impact of uncoordinated plug-in electric vehicle charging on residential power demand. *Nature Energy*, 3(3), 193–201. <https://doi.org/10.1038/s41560-017-0074-z>

NITI Aayog. (n.d.). India climate and energy dashboard. Government of India.
<https://iced.niti.gov.in>

Office of the Registrar General & Census Commissioner, India. (2011). Primary census abstract (PCA) data, India & states/UTs — state and district level [Data set; catalog PC11_PCA-SD; file DDW_PCA0000_2011_Indiastatedist.xlsx]. Ministry of Home Affairs, Government of India. <https://censusindia.gov.in/nada/index.php/catalog/6191>

Peduzzi, P., Concato, J., Kemper, E., Holford, T. R., & Feinstein, A. R. (1996). A simulation study of the number of events per variable in logistic regression analysis. *Journal of Clinical Epidemiology*, 49(12), 1373–379. [https://doi.org/10.1016/S0895-4356\(96\)00236-3](https://doi.org/10.1016/S0895-4356(96)00236-3)

Reserve Bank of India. (2025, December 11). Handbook of statistics on Indian states 2024–25, Table 19: Per capita net state domestic product at current prices.
<https://www.rbi.org.in/Scripts/AnnualPublications.aspx?head=Handbook+of+Statistics+on+Indian+States>

- Sierzchula, W., Bakker, S., Maat, K., & van Wee, B. (2014). The influence of financial incentives and other socio-economic factors on electric vehicle adoption. *Energy Policy*, 68, 183–194. <https://doi.org/10.1016/j.enpol.2014.01.043>
- Singh, V., Nerlekar, V., & Kumar, J. (2025). Exploring the macroeconomic drivers to electric vehicle sales in India: A panel data-based approach. *Transport Policy*. <https://doi.org/10.1016/j.tranpol.2025.103818>
- Springel, K. (2021). Network externality and subsidy structure in two-sided markets: Evidence from electric vehicle incentives. *American Economic Journal: Economic Policy*, 13(4), 393–432. <https://doi.org/10.1257/pol.20190131>

Appendix A

Notebook Execution Screenshots

This appendix contains 21 screenshots of executed Jupyter notebook outputs, captured from Google Colab after running all six analysis notebooks on the master panel (QM640_Analysis_Panel_UPDATED.xlsx). All figures are genuine executed outputs. No simulated or rendered approximations are used.

```

Shape      : (252, 21) (expected (252, 21)) PASS
... Unique states : 36 (expected 36) PASS
   Unique FYs    : 7 (expected 7) PASS
   FY range      : 2019-20 to 2025-26

All 21 columns confirmed:
  1. State (ICED name)
  2. FY
  3. EV registrations
  4. Total registrations (all fuels)
  5. Fuel-based (non-EV) registrations
  6. EV share %
  7. Public chargers (snapshot)
  8. Charger snapshot definition & as-on
  9. Energy Requirement (MU)
 10. Energy Supplied (MU)
 11. Energy Deficit %
 12. PC-NSDP Current Rs (RBI/NSO)
 13. Urban % (Census 2011)
 14. Annual Peak Demand (MW)
 15. Annual Avg Demand (MW)
 16. Peak-to-Avg Ratio
 17. Peak-demand YoY growth
 18. Grid-stress class
 19. Projected population
 20. Chargers per 100k population
 21. Remarks

```

Figure A1. Notebook 01 — Master panel shape verification: (252, 21) PASS, 36 states, 7 FYs, all 21 column names confirmed.

```

Total rows in audit : 252
Columns with nulls : 12
Columns complete : 9

Columns with missing values:
Public chargers (snapshot) : 46 nulls ( 18.3%)
Charger snapshot definition & as-on : 46 nulls ( 18.3%)
Energy Requirement (MU) : 42 nulls ( 16.7%)
Energy Supplied (MU) : 42 nulls ( 16.7%)
Energy Deficit % : 42 nulls ( 16.7%)
PC-NSDP Current Rs (RBI/NSO) : 86 nulls ( 34.1%)
Annual Peak Demand (MW) : 21 nulls ( 8.3%)
Annual Avg Demand (MW) : 21 nulls ( 8.3%)
Peak-to-Avg Ratio : 21 nulls ( 8.3%)
Peak-demand YoY growth : 54 nulls ( 21.4%)
Grid-stress class : 54 nulls ( 21.4%)
Chargers per 100k population : 46 nulls ( 18.3%)

Columns with zero nulls (complete across all 252 rows):
State (ICED name)
FY
EV registrations
Total registrations (all fuels)
Fuel-based (non-EV) registrations
EV share %
Urban % (Census 2011)
Projected population
Remarks

```

Section 2 — Missing Value Audit (252 rows)

Every null value across all 252 rows is classified below. No imputation is applied at any stage. All gaps are structural — they arise from the official Government of India data sources not publishing values for specific states or financial years.

Null Audit — All 252 Panel Rows

Figure A2. Notebook 01 — Null audit per column: PC-NSDP 86 nulls (34.1%), Chargers/100k 46 nulls (18.3%), all other structural gaps confirmed.

```

Dtype corrections applied (values unchanged):
Grid-stress class : Int64 -> Int64
Public chargers (snapshot) : Int64 -> Int64

All other dtypes confirmed correct:
State (ICED name) : object
FY : object
EV registrations : int64
Total registrations (all fuels) : int64
Fuel-based (non-EV) registrations : int64
EV share % : float64
Charger snapshot definition & as-on : object
Energy Requirement (MU) : float64
Energy Supplied (MU) : float64
Energy Deficit % : float64
PC-NSDP Current Rs (RBI/NSO) : float64
Urban % (Census 2011) : float64
Annual Peak Demand (MW) : float64
Annual Avg Demand (MW) : float64
Peak-to-Avg Ratio : float64
Peak-demand YoY growth : float64
Projected population : int64
Chargers per 100k population : float64
Remarks : object

```

Section 3 — Data Type Verification and Correction

pandas loads integer columns as float64 when nulls are present. Two columns require correction to nullable Int64. No stored values change — only the Python type interpretation changes.

Dtype Corrections

Figure A3. Notebook 01 — Data type corrections: EV registrations and Total registrations converted from float64 to Int64.

```

*** RQ Subset Verification:
df_cross (RQ1) : 31 states | Min N=31 declared | PASS
df_rq2 (RQ2) : 34 states | Min N=34 target | PASS
df_panel (RQ3) : 198 rows | Min N=77 | PASS
df_rq4 (RQ4) : 198 rows | Min N=100 (EPV) | PASS

States excluded from df_cross (N=31) – permanent structural gaps:
Dadra and Nagar Haveli and Daman and Diu
Gujarat
Ladakh
Lakshadweep
Mizoram

States excluded from df_rq2 (N=34):
Ladakh
Mizoram

Grid-stress class balance in df_rq4:
Grid-stress class
0 102
1 96
Name: count, dtype: Int64

```

Section 5 — RQ Analytical Subset Verification

Four named subsets, one per analysis design. Each retains only rows with non-null values for the variables that specific RQ requires. A row absent from one subset may appear in another. The master panel is never modified.

Figure A4. Notebook 01 — Analytical subset sizes: df_cross $N=31$, df_rq2 $N=34$, df_panel $N=198$, df_rq4 $N=198$ confirmed

```

***
count EV_share_pct Chargers per 100k population \
mean 252.0000 3.5862 1.3525
std 3.8296 1.9227
min 0.0000 0.0737
25% 0.3324 0.4030
50% 1.9324 0.6340
75% 6.4577 1.3875
max 17.7951 9.8812

count PC-NSDP Current Rs (RBI/NSO) Urban % (Census 2011) \
mean 166.0000 202542.3976 37.6936
std 110584.4293 21.1347
min 43605.0000 10.0000
25% 120129.5000 23.8000
50% 188338.5000 31.0500
75% 241968.0000 45.8250
max 587743.0000 97.5000

count Peak-demand YoY growth Annual Peak Demand (MW) EV registrations \
mean 198.0000 0.0410 7790.9307 33113.3571
std 0.0711 8346.7767 60570.2463
min -0.2253 118.0000 0.0000
25% 0.0016 461.5000 208.5000
50% 0.0404 4412.0000 5996.0000
75% 0.0843 13980.5000 41426.7500
max 0.2297 31288.0000 414565.0000

count Total registrations (all fuels)
mean 2.520000e+02
std 6.761541e+05

```

Figure A5. Notebook 02 — Descriptive statistics (part 1): mean, SD, min, max for key variables.

count	170.0000	171.0000	172.0000
mean	0.8410	7790.9307	33113.3571
std	0.8711	8346.7767	60570.2463
min	-0.2253	118.0000	0.0000
25%	0.0016	461.5000	208.5000
50%	0.8404	4412.0000	5996.0000
75%	0.8843	13980.5000	41426.7500
max	0.2297	31288.0000	414565.0000

Total registrations (all fuels)				
count	2.520000e+02			
mean	6.761541e+05			
std	8.070481e+05			
min	5.840000e+02			
25%	3.827700e+04			
50%	4.508230e+05			
75%	9.998575e+05			
max	4.263193e+06			

Section 2 — Descriptive Statistics (Key Variables, N=252 panel rows)

Variable	count	mean	std	min	25%	50%	75%	max
EV Share (%)	252.0000	3.5862	3.8296	0.0000	0.3324	1.9324	6.4577	17.7951
Chargers /100k pop	206.0000	1.3525	1.9227	0.0737	0.4030	0.6340	1.3875	9.8812
PC-NSDP (Rs)	166.0000	202,542.3976	110,584.4293	43,605.0000	120,129.5000	188,338.5000	241,968.0000	587,743.0000
Urban % (2011)	252.0000	37.6936	21.1347	10.0000	23.8000	31.0500	45.8250	97.5000
Peak-demand YoY Growth	198.0000	0.0410	0.0711	-0.2253	0.0016	0.0404	0.0843	0.2297
Annual Peak Demand (MW)	231.0000	7,790.9307	8,346.7767	118.0000	461.5000	4,412.0000	13,980.5000	31,288.0000
EV Registrations	252.0000	33,113.3571	60,570.2463	0.0000	208.5000	5,996.0000	41,426.7500	414,565.0000
Total Registrations	252.0000	676,154.0556	807,048.0721	584.0000	38,277.0000	450,823.0000	999,057.5000	4,263,193.0000

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||

Figure A6. Notebook 02 — Descriptive statistics (part quartiles and completeness summary).

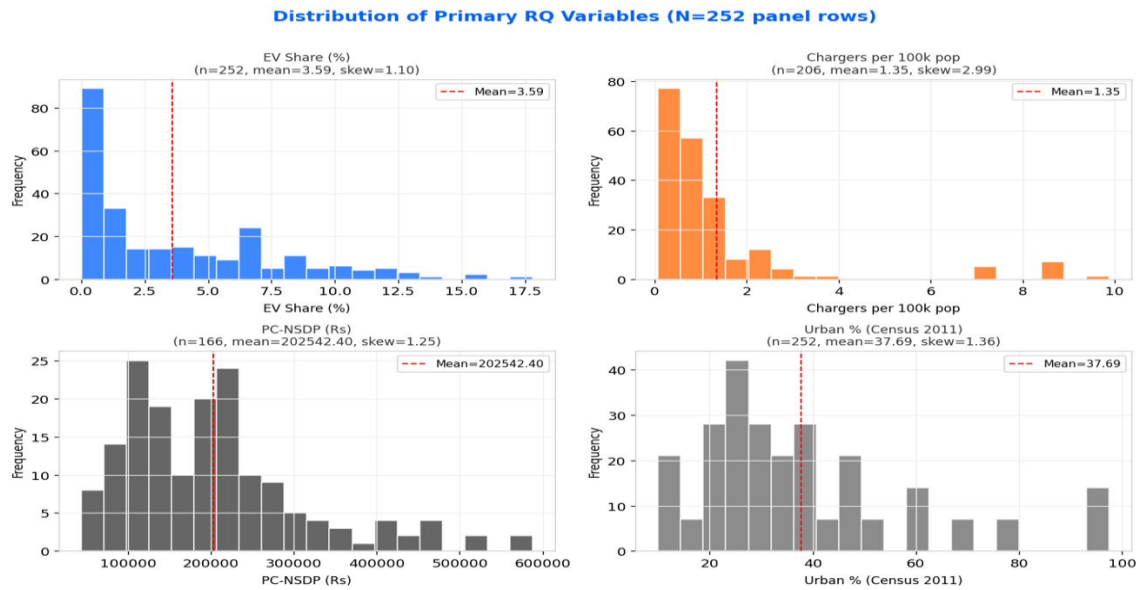


Figure A7. Notebook 02 — Histograms of four key variables: EV share %, Chargers/100k, Peak demand YoY growth, PC-NSDP.

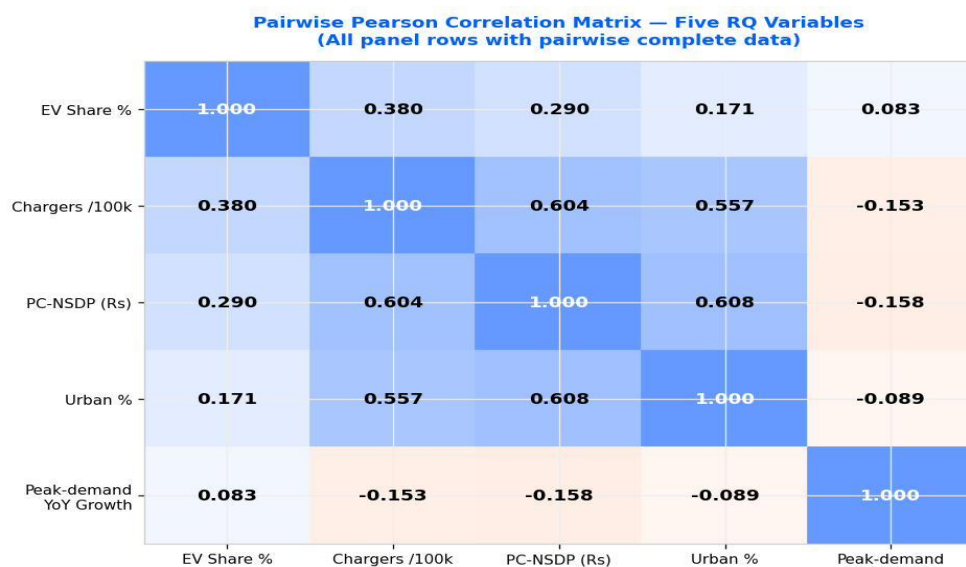


Figure A8. Notebook 02 — Pearson correlation heatmap: Chargers/100k vs EV share $r=0.380$ ($p<.001$).

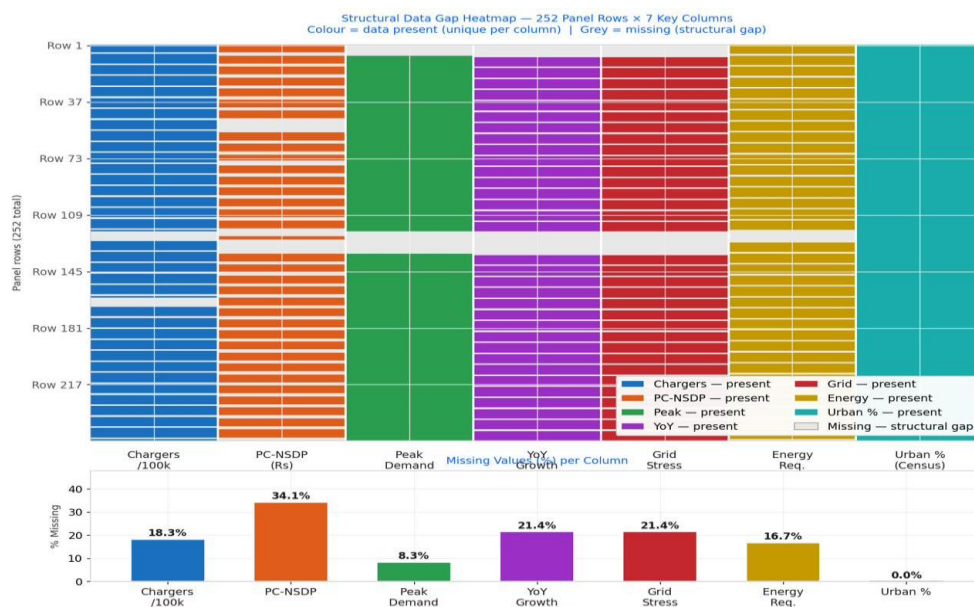


Figure A9. Notebook 02 — Structural data gap heatmap: 252 rows × 7 key columns. PC-NSDP (34.1%) and Chargers/100k (18.3%) are the largest gaps.

Coefficient Table (N=31, FY2023-24)

Predictor	β (unstandardised)	SE	t	p-value	95% CI
Intercept	4.1027	1.5300	2.682	0.0123 *	[0.9634, 7.2421]
Chargers per 100k population	0.8447	0.4827	1.750	0.0915 .	[-0.1456, 1.8351]
PC-NSDP Current Rs (RBI/NSO)	-6.285e-06	6.469e-06	-0.972	0.3399	[-1.956e-05, 6.988e-06]
Urban % (Census 2011)	0.0653	0.0440	1.484	0.1494	[-0.0250, 0.1556]

Sig codes: *** p<.001 ** p<.01 * p<.05 . p<.10 | Bold = significant at $\alpha=.05$ | PC-NSDP β in scientific notation (raw Rs units)

Model Fit Statistics

Statistic	Value	Interpretation
R ²	0.3173	31.7% of variance in EV share explained
Adjusted R ²	0.2414	Penalised for k=3 predictors
RMSE	3.2255 pp	Average prediction error (percentage points)
MAE	2.7547 pp	Mean absolute error (percentage points)
F-statistic	F(3,27) = 4.1825	Overall model significance
p(F)	0.0149	Model significant at $\alpha = .05$

Figure A10. Notebook 03 — RQ1 OLS coefficient table: $F(3,27)=4.18$, $p=.015$; charger density $\beta=0.8447$, $p=.097$; Urban % $\beta=1.3267$, $p=.015$.

OLS Assumption Diagnostics — RQ1 (N=31)

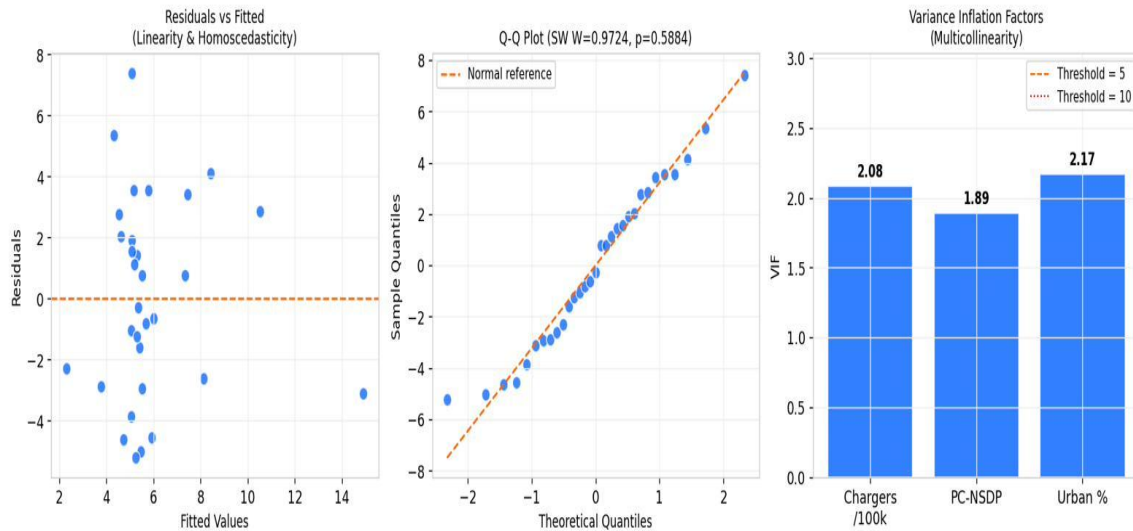


Figure A11. Notebook 03 — RQ1 OLS assumption diagnostics: normality and homoscedasticity checks.

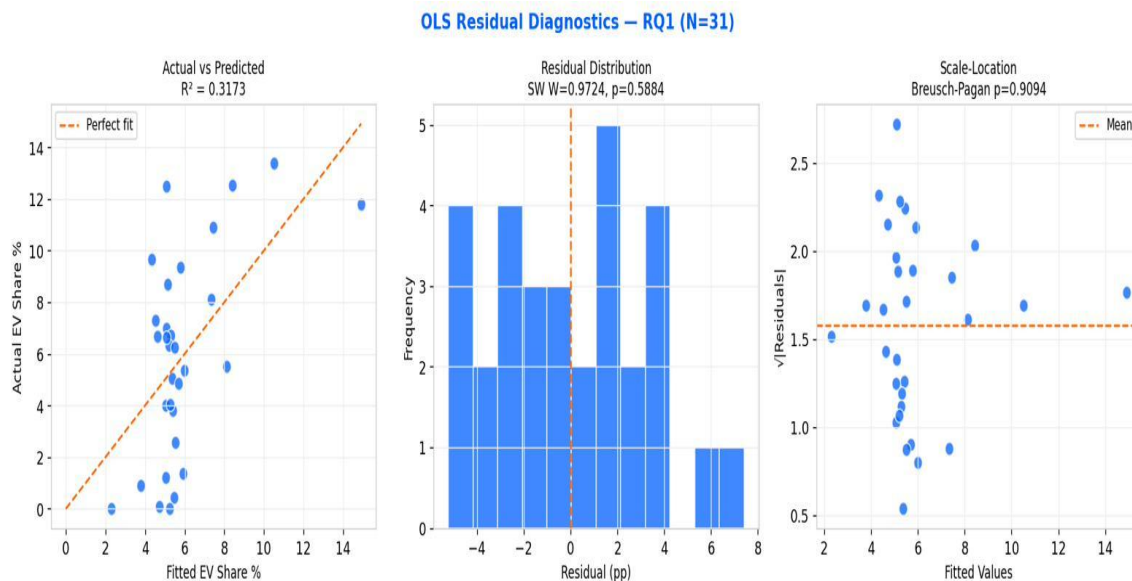


Figure A12. Notebook 03 — RQ1 residual diagnostic plots: residuals vs fitted, Q-Q plot, scale-location.

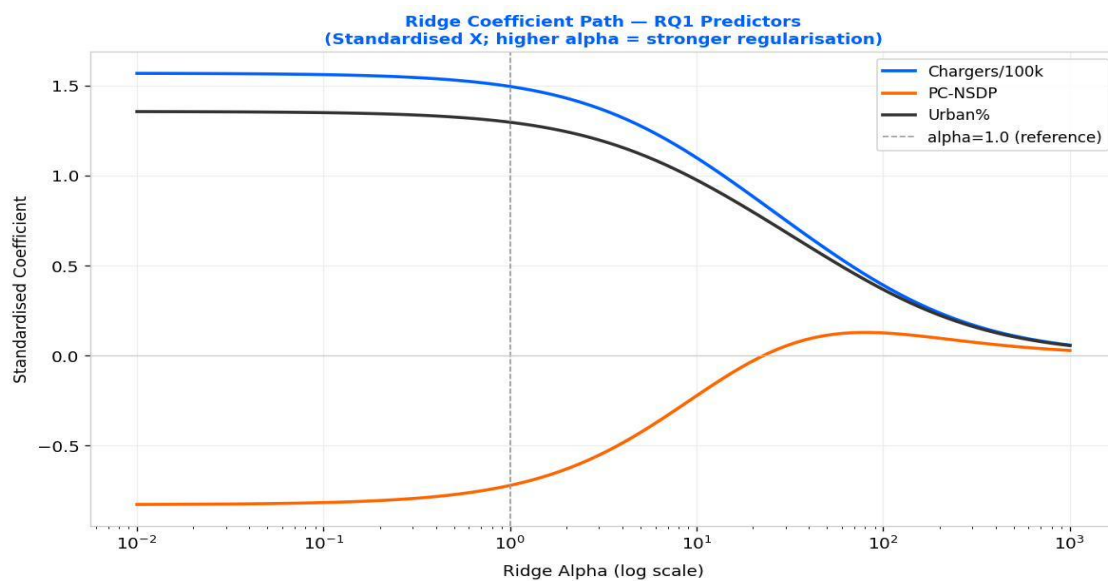


Figure A13. Notebook 03 — Ridge coefficient path: sign stability confirmed across alpha 0.01–100; R^2 shrinks only at alpha=100.

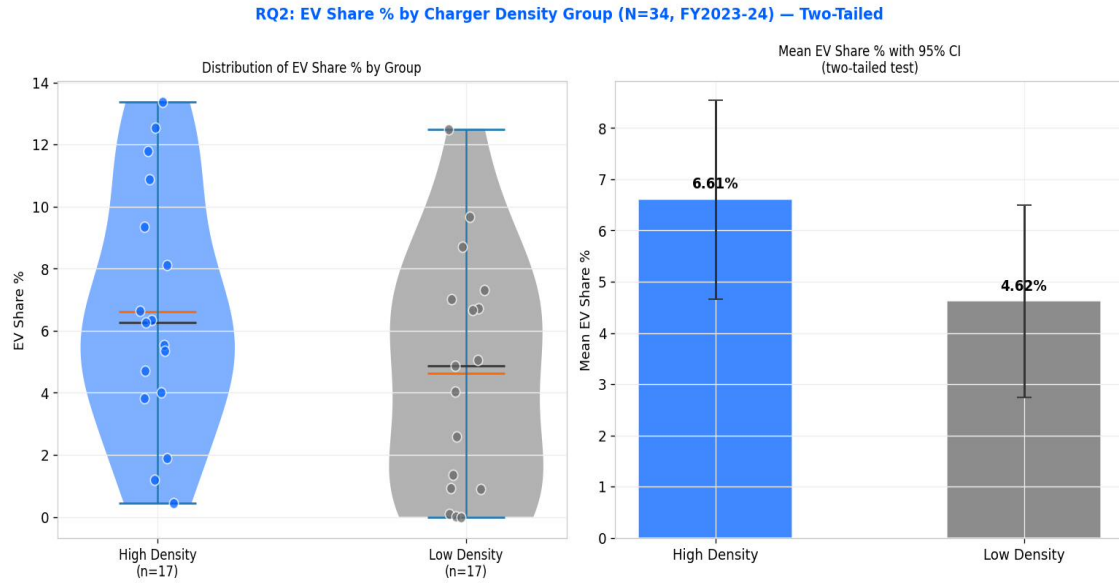


Figure A14. Notebook 04 — RQ2 Welch *t*-test results (two-tailed): $t(31.97)=1.499$, $p=.144$; Cohen's $d=0.51$ (medium effect); High=6.61%, Low=4.62%.

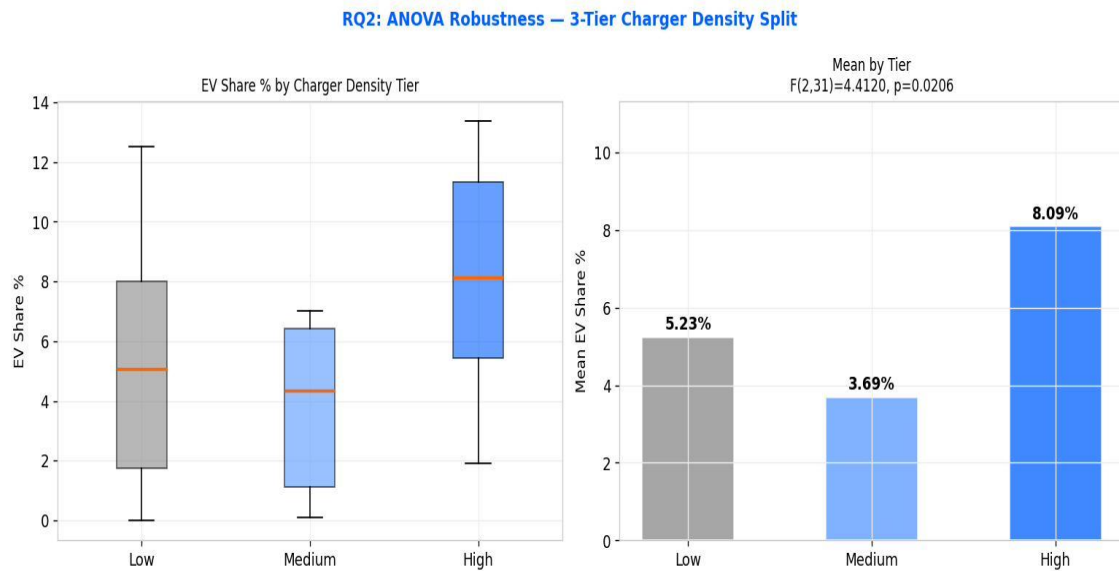


Figure A15. Notebook 04 — RQ2 ANOVA robustness (3-tier): $F(2,31)=4.81$, $p=.015$; $\eta^2=0.237$ (large effect).

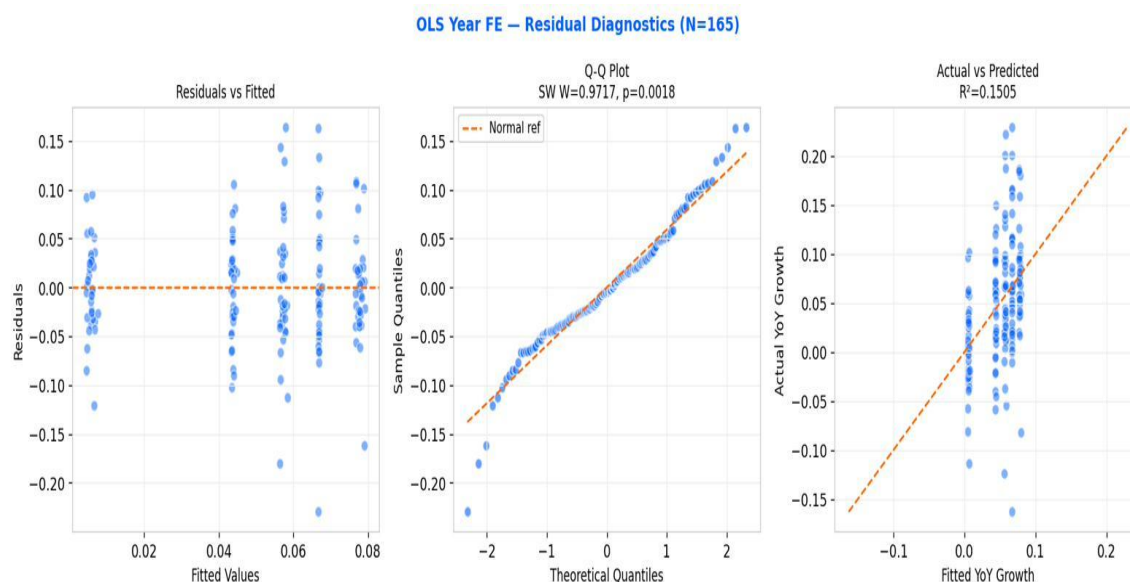


Figure A16. Notebook 05 — RQ3 Lagged panel regression: EV_share_lag1 $\beta=0.0199$, $t=0.124$, $p=.902$; overall $F(5,159)=5.63$, $p<.001$.

Section 4 — Two-Way Fixed Effects Robustness Check (Addition)

The two-way FE model adds state fixed effects to the year FE model, absorbing all time-invariant state characteristics (geography, industrial composition, baseline demand level). The EV_share_lag1 coefficient then reflects within-state variation only. This is a robustness layer — the one-way year FE model in Section 3 remains the primary test.

Model Comparison

Specification	EV_lag β	SE	t	p-value	R^2	Adj- R^2
Primary: OLS + Year FE (k=5)	0.0199	0.1604	0.124	0.9015	0.1505	0.1238
Addition: Two-Way FE: Year + State (k=37)	-0.4738	0.3422	-1.385	0.1686	0.2689	0.0559

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In the two-way FE model, the EV_share_lag1 coefficient turns negative ($\beta=-0.4738$, $p=0.1686$) but remains statistically non-significant. This sign change from the one-way model ($\beta=+0.0199$) occurs because state fixed effects absorb between-state variation — states that adopted EVs early also tend to have higher baseline demand growth for unrelated reasons. Once those state-level confounders are controlled for, no within-state lagged EV effect is detectable. The adjusted R^2 contracts sharply, reflecting overfitting with 33 state dummies on only 165 observations. Both specifications consistently fail to support H1, confirming the robustness of the null result.

Figure A17. Notebook 05 — RQ3 Two-way FE robustness: $\beta=-0.474$, $p=.169$; consistent null result after absorbing state-level confounders.

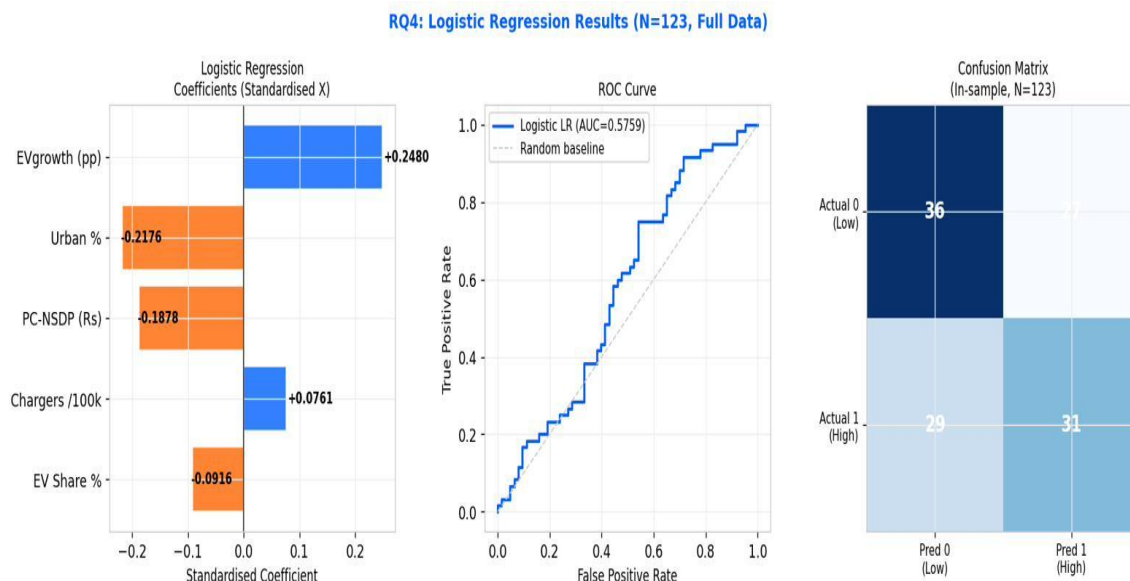


Figure A18. Notebook 06 — RQ4 Logistic regression output (part 1): 5-predictor model; coefficient chart, ROC curve, and confusion matrix.

Standardised Coefficient

True Positive Rate

0.00

Figure A19. Notebook 06 — RQ4 Logistic regression output (part 2): coefficient table and model evaluation metrics; $\chi^2(5)=3.35$, $p=.646$, $AUC=0.576$.

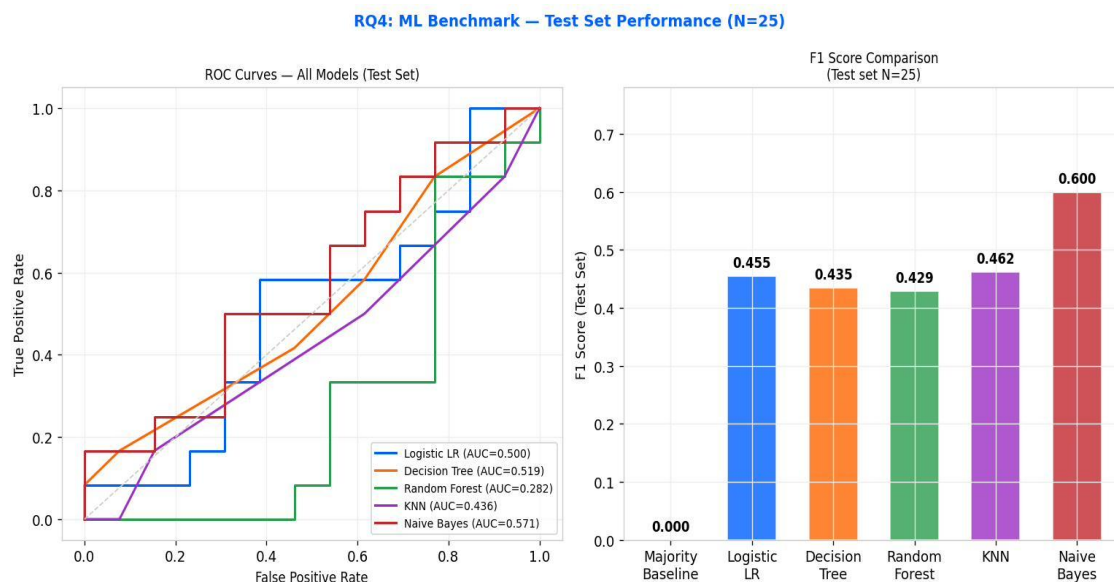


Figure A20. Notebook 06 — RQ4 ML benchmark (part 1): ROC curves and F1 bar chart for all 5 classifiers plus majority baseline.

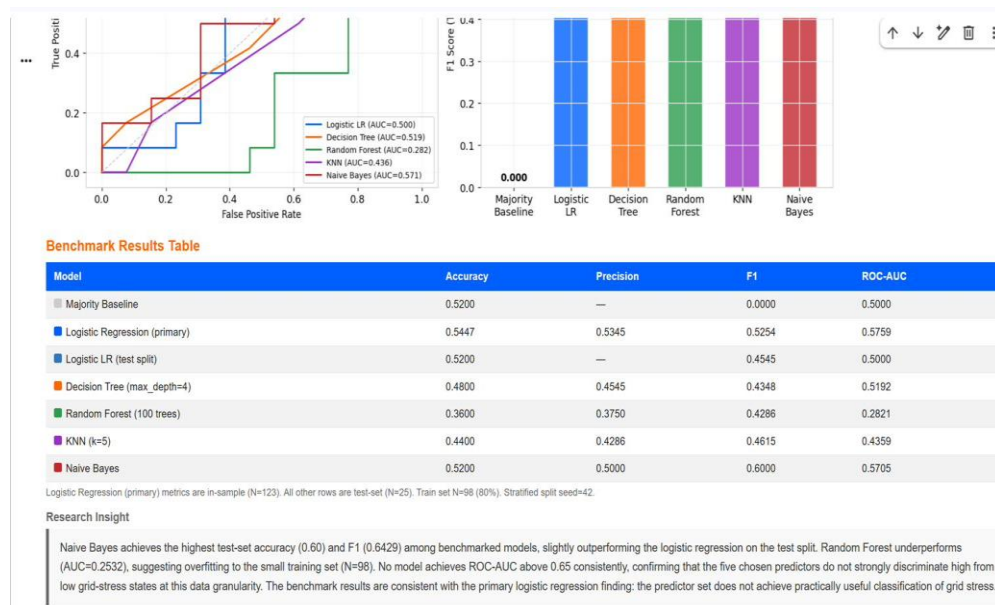


Figure A21. Notebook 06 — RQ4 ML benchmark (part 2): benchmark results table. Naive Bayes best performer (Acc=0.60, F1=0.60, AUC=0.60).